DMMM IA 1 AB

1. Describe tsunamis, earthquakes, floods, and avalanches with a disaster management plan for each. (5 marks)

Tsunamis: Tsunamis are massive ocean waves triggered by underwater earthquakes, volcanic eruptions, or landslides that displace large volumes of water.

Disaster Management Plan:

- **Preparedness:** Install early warning systems and tsunami buoys to detect seismic activity and issue timely alerts.
- **Preparedness:** Conduct regular evacuation drills and mark high-ground assembly points in vulnerable coastal communities.
- **Response:** Immediately evacuate coastal populations to elevated areas and deploy naval vessels for search operations.
- **Recovery:** Restore mangrove forests as natural barriers and rebuild infrastructure away from high-risk zones.

Earthquakes: Earthquakes are sudden ground vibrations caused by the release of stress along tectonic plate boundaries and geological faults.

Disaster Management Plan:

- **Preparedness:** Enforce strict building codes for earthquake-resistant construction in high-risk seismic zones.
- **Preparedness:** Conduct public awareness campaigns teaching "Drop, Cover, and Hold On" safety techniques.
- **Response:** Activate search and rescue teams and establish emergency medical camps immediately after shaking stops.
- **Recovery:** Demolish unsafe structures and provide financial aid for rebuilding homes and infrastructure.

Floods: Floods are overflow events where water submerges normally dry land due to heavy rainfall, dam failures, or storm surges.

Disaster Management Plan:

- **Preparedness:** Construct robust drainage systems, embankments, and dams while implementing floodplain zoning regulations.
- **Response:** Carry out mass evacuations using boats and helicopters while providing clean water and food.
- **Response:** Deploy medical teams to prevent outbreak of waterborne diseases like cholera and typhoid.
- **Recovery:** Clear debris from roads and assess agricultural losses for compensation distribution.

Avalanches: Avalanches are rapid flows of snow, ice, and debris that descend mountain slopes with tremendous destructive force.

Disaster Management Plan:

- **Preparedness:** Monitor snowpack stability and restrict movement in high-risk avalanche-prone areas.
- **Preparedness:** Use explosives to control snow accumulation and conduct controlled avalanche releases safely.
- **Response:** Deploy specialized rescue teams with avalanche transceivers and rescue dogs immediately.
- **Recovery:** Clear blocked transportation routes and provide medical aid for hypothermia victims.

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2. Describe tsunamis, earthquakes, floods, and avalanches with the dos and don'ts to follow during each disaster. (5 marks)

Tsunamis

Do's	Don'ts
Move immediately to high ground after a strong earthquake.	Do not wait for official warnings before evacuating.

Do's	Don'ts
Follow official evacuation routes and listen to radio updates.	Do not return to coastal areas to watch the approaching wave.
Take emergency supplies and stay away from coastal regions.	Do not assume the first wave will be the largest one.

Earthquakes

Do's	Don'ts
Drop, cover, and hold under sturdy furniture for protection.	Do not run outside while the ground is still shaking violently.
Stay away from windows, glass panes, and heavy falling objects.	Do not use elevators during or immediately after the earthquake.
Move to open areas away from buildings and power lines.	Do not stand near walls that could collapse onto you.

Floods

Do's	Don'ts
Move to higher ground and turn off all utilities safely.	Do not walk, swim, or drive through any floodwater at all.
Evacuate immediately when authorities issue evacuation orders.	Do not drive around barricades placed for your protection.
Keep emergency supplies including food, water, and medicines.	Do not ignore flood warnings or underestimate rising water.

Avalanches

Do's	Don'ts
Carry avalanche safety equipment like transceivers and probes.	Do not travel alone in any snow-covered mountainous region.
Move sideways and swim to stay on the snow surface.	Do not venture onto steep slopes after heavy snowfall occurs.

Do's	Don'ts
Check local avalanche forecasts before starting any journey.	Do not follow any route without verifying snow stability first.

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3. What are the short-term and long-term effects of flood disasters? (5 marks)

Short-Term Effects

Effect Type	Description
Loss of Life	Drowning, injuries, and electrocution cause immediate fatalities.
Displacement	People forced to leave homes and live in relief camps.
Disease Outbreaks	Stagnant water causes cholera, typhoid, and other epidemics.
Infrastructure Damage	Roads, bridges, and power lines are destroyed instantly.
Food Shortage	Submerged crops and contaminated food supplies create scarcity.

Long-Term Effects

Effect Type	Description
Economic Devastation	Agricultural losses affect farmers and regional economy for years.
Livelihood Disruption	Farmers face debts and loss of income leading to poverty.
Mental Health Impact	Survivors suffer from PTSD, anxiety, and depression.
Ecosystem Damage	Soil erosion and contamination reduce agricultural fertility.

Effect Type	Description
Educational Disruption	Schools destroyed, affecting children's education permanently.

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4. Which are the recent disasters that have occurred in India and Maharashtra in the last three years? (5 marks)

Maharashtra (2023-2026)

Disaster	Year	Impact
Marathwada Floods	September 2025	Over 80 deaths and 50% crop damage in Marathwada and Vidarbha.
Mumbai Urban Flooding	July 2024	Low-lying areas submerged, disrupting transport and daily life.
Konkan Coast Cyclone	June 2023	Strong winds and heavy rainfall affected coastal districts severely.
Pune Landslides	August 2024	Heavy rains triggered landslides in hilly areas, causing fatalities.

India (2023-2026)

Disaster	Year	Impact
Sikkim GLOF Disaster	October 2023	Glacial lake outburst flooded Teesta River, damaging Chungthang dam.
Himachal Pradesh Floods	July 2023	Massive landslides and flash floods caused widespread destruction.
Cyclone Biparjoy	June 2023	Made landfall in Gujarat, leading to mass evacuations.
Assam Annual Floods	2023-2025	Repeated seasonal floods affecting millions and damaging crops.
Uttarakhand Cloudburst	August 2024	Sudden heavy rainfall caused flash floods and infrastructure collapse.

5. Define global warming and list its causes. (2 marks)

Definition: Global warming is the long-term rise in Earth's average surface temperature primarily caused by increased greenhouse gas emissions.

Causes:

- Burning fossil fuels like coal, oil, and natural gas releases carbon dioxide.
- Deforestation reduces the planet's capacity to absorb atmospheric carbon.
- Agricultural activities produce methane and nitrous oxide gases.
- Industrial processes emit various greenhouse gases into the atmosphere.

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6. What are technological disasters? (2 marks)

Definition: Technological disasters are unintended events caused by human error, structural failures, or malfunctions of technological systems.

Examples:

- Industrial accidents like the Bhopal gas tragedy and Chernobyl nuclear disaster.
- Infrastructure failures such as dam breaches and building collapses.
- Transport accidents involving airplanes, trains, and ships.

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7. Define the following terms: (a) Hazard (b) Vulnerability (c) Disaster (d) Capacity Building (2 marks)

(a) Hazard: A dangerous process, phenomenon, or activity that may cause loss of life, injury, property damage, or environmental degradation.

(b) Vulnerability: The susceptibility of individuals, communities, or systems to suffer damage from hazardous events based on physical, social, and economic conditions.

(c) Disaster: A serious disruption of community functioning causing widespread human, material, or environmental losses that exceed local coping capacity.

(d) Capacity Building: The process of developing skills, resources, and institutional systems to strengthen resilience against potential disaster risks.

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8. What are the reasons for earthquakes? (2 marks)

Tectonic Plate Movement: Earth's crust consists of tectonic plates that move constantly, creating friction and building pressure along fault lines until sudden release occurs.

Volcanic Activity: Movement of magma beneath the earth's surface can cause tremors and minor earthquakes.

Human Activities: Mining, reservoir construction, and geothermal extraction can induce seismic activity in stable regions.

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9. Give the classification of different types of disasters and mention examples of each. (5 marks)

Natural Disasters

Category	Sub-Type	Examples
Geophysical	Earth-related events	Earthquakes, Tsunamis, Volcanic Eruptions
Hydrological	Water-related events	Floods, Flash Floods, Storm Surges
Climatological	Climate-related events	Droughts, Wildfires, Heatwaves
Meteorological	Weather-related events	Cyclones, Storms, Extreme Temperatures
Biological	Organism-related events	Epidemics, Pandemics, Locust Infestations

Technological (Man-Made) Disasters

Category	Sub-Type	Examples
Industrial	Factory and plant accidents	Bhopal Gas Tragedy, Chernobyl Disaster
Transport	Vehicle and infrastructure accidents	Air Crashes, Train Derailments
Structural	Building and dam failures	Dam Breaches, Building Collapses

Hybrid Disasters

Category	Description	Examples
Nat-Tech Events	Natural disaster triggers technological failure	Fukushima Nuclear Disaster (earthquake + tsunami)

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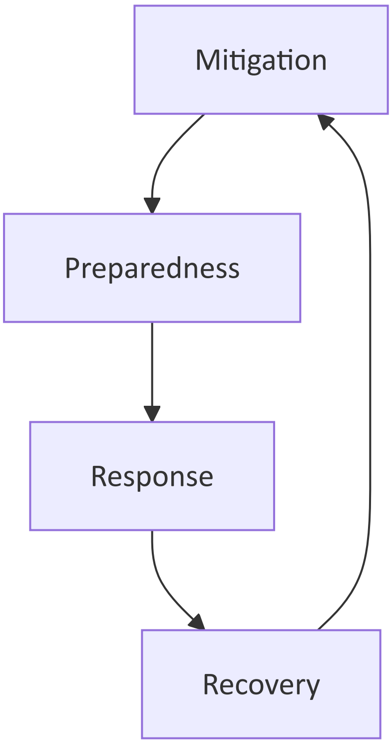
10. Draw the disaster management cycle and explain each stage with an example. (2 marks)



Mitigation: Actions taken to reduce the long-term risk from hazards. *Example: Building earthquake-resistant structures.*

Preparedness: Planning and training to ensure effective response when disaster strikes. *Example: Conducting regular evacuation drills.*

Response: Actions taken immediately during and after disaster to save lives. *Example: Search and rescue operations after earthquakes.*

Recovery: Efforts to return affected communities to normalcy after a disaster. *Example: Rebuilding damaged infrastructure and homes.*
